plentiful nutrients. In summer, these oceans provide rich pickings for the many marine mammals and seabirds that migrate to these areas to feed and breed. The tundra also has summer visitors, such as reindeer that spend the winter sheltering from the cold in the taiga forest to the south, but return each summer.

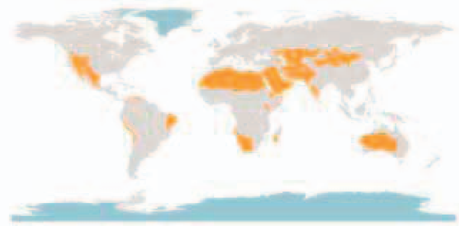
Desert regions

Most of the world's great hot deserts, such as Africa's Sahara, are found in the subtropics, where dry conditions persist for months at a time. Others, such as the Mojave in the US southwest, are found on the dry leeward side of mountains. A few, such as the Atacama Desert in South America, are coastal and lack rain due to cold offshore water inhibiting cloud formation. Cold deserts are found in continental interiors and are very hot in summer and very cold in winter. All of these deserts are dry, receiving less than 6 in (15 cm) rain annually, and most are cloudless. The exception is coastal desert, which can benefit from early morning fog drifting in from the ocean. Hot deserts, although hot all year, are cold at night as the lack of cloud cover allows heat to escape freely. Cold deserts also have large daily temperature fluctuations, but in winter, temperatures are below freezing and snow is not uncommon.

Plants growing in deserts must cope not only with lack of water and extreme temperatures, but also with soils that have little organic matter and few soil microorganisms. All desert plants and animals tend to restrict their reproductive efforts to periods of rainfall and show a range of adaptations to heat, such as being able to retain or store water, and many animals forage at night.

DESERT AND POLAR ICE DISTRIBUTION

Deserts in the southern hemisphere tend to be less extensive than those in the north, which include the largest desert of all: the Sahara. Almost all of Antarctica and most of Greenland in the Arctic Circle is covered in ice-sheets.



Desert   Ice

COLD DESERT

Despite the hostile treeless environment, limited rainfall, and enormous seasonal temperature differences, cold deserts are home to a variety of animals. Small mammals include the dwarf hamster, large ones the critically endangered Gobi bear.

COASTAL DESERT

In the early morning, coastal deserts benefit from moisture carried inland as fog. It is an important source of water for a number of arthropods and reptiles, some of which have adaptations to enhance water collection and storage.

HOT DESERT

Daytime temperatures in hot deserts are so high that even "cold blooded" arthropods, which are reliant on the Sun for warmth, seek shade to avoid the heat. Many animals, such as the deathstalker scorpion, avoid the Sun completely by being nocturnal.